

PL/SQL Programming

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Exercise 1: Control Structures

Scenario 1: Discount loan interest for customers above 60

```
DECLARE

    v_age NUMBER;

BEGIN

    FOR c IN (SELECT CustomerID, DOB FROM Customers) LOOP

        v_age := FLOOR(MONTHS_BETWEEN(SYSDATE, c.DOB)/12);

        IF v_age > 60 THEN

            UPDATE Loans

            SET InterestRate = InterestRate - 1

            WHERE CustomerID = c.CustomerID;

        END IF;

    END LOOP;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Interest rates updated');

END;
```

Scenario 2: Set VIP status for customers with balance > 10000

```
ALTER TABLE Customers

ADD IsVIP VARCHAR2(5);
```

```

BEGIN
    FOR c IN (SELECT CustomerID, Balance FROM Customers) LOOP

        IF c.Balance > 10000 THEN

            UPDATE Customers

            SET IsVIP='TRUE'

            WHERE CustomerID=c.CustomerID;

        END IF;

    END LOOP;

    COMMIT;
END;

```

Scenario 3: Loan due within next 30 days

```

BEGIN
    FOR I IN
    (
        SELECT CustomerID, LoanID, EndDate
        FROM Loans
        WHERE EndDate BETWEEN SYSDATE AND SYSDATE+30
    )
    LOOP
        DBMS_OUTPUT.PUT_LINE(
            'Reminder: Customer '
            || I.CustomerID ||
            ' Loan '
            || I.LoanID ||
            ' due on '
            || I.EndDate);
    END LOOP;
END;

```

Exercise 3: Stored Procedures

Scenario 1: ProcessMonthlyInterest

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
IS
BEGIN

    UPDATE Accounts
    SET Balance = Balance + (Balance*0.01)
    WHERE AccountType='Savings';

    COMMIT;

END;
```

Scenario 2: UpdateEmployeeBonus

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
    p_department VARCHAR2,
    p_bonus NUMBER
)
IS
BEGIN

    UPDATE Employees
    SET Salary = Salary + (Salary*p_bonus/100)
    WHERE Department=p_department;

    COMMIT;

END;
```

Scenario 3: TransferFunds

```
CREATE OR REPLACE PROCEDURE TransferFunds(
    p_from NUMBER,
    p_to NUMBER,
    p_amount NUMBER
)
```

```
IS
  v_balance NUMBER;
BEGIN

  SELECT Balance
  INTO v_balance
  FROM Accounts
  WHERE AccountID=p_from;

  IF v_balance >= p_amount THEN

    UPDATE Accounts
    SET Balance=Balance-p_amount
    WHERE AccountID=p_from;

    UPDATE Accounts
    SET Balance=Balance+p_amount
    WHERE AccountID=p_to;

    COMMIT;

  ELSE
    DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
  END IF;

END;
```